Laura H. Yang

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EDUCATION

Harvard University — Cambridge, MA
Ph.D., Environmental Science and Engineering — 09/2021–08/2025

Georgia Institute of Technology — Atlanta, GA
B.S., Environmental Engineering — 09/2016–12/2020

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow, Washington University in St. Louis — 09/2025–Present
Host: Randall V. Martin

NSF Graduate Research Fellow, Harvard University — 09/2021–08/2025
Advisor: Daniel J. Jacob

NSF Intern Program Research Intern, NASA Jet Propulsion Laboratory — 02/2025–07/2025
Hosts: Kevin W. Bowman, Kazuyuki Miyazaki

Undergraduate Research Assistant, Georgia Institute of Technology — 05/2019–08/2021
Advisor: Nga L. (Sally) Ng

HONORS, AWARDS, AND FELLOWSHIPS

Selected Participant, EAS Trailblazers Symposium, Caltech	2026
Air Quality and Health Initiative Postdoctoral Fellowship, Washington University in St. Louis	2025
Intern Supplemental Funding, National Science Foundation	2025
Graduate Research Fellowship, National Science Foundation	2021–2025
Stonington Endowment Graduate Fellowship, Harvard University	2021
Sigma Xi Best Undergraduate Research Award, Georgia Tech	2021
Civil and Environmental Engineering Best Undergraduate Research Award, Georgia Tech	2021
Buck Stith Outstanding Senior Award, Georgia Tech	2021
President's Undergraduate Research Awards, Georgia Tech	2017, 2019
Buck Stith Outstanding Junior Award, Georgia Tech	2019
Callaway Scholarship, Georgia Tech	2017
Provost Scholarship, Georgia Tech	2016–2020
Outstanding Young Environmental Scientist Award, Cary Institute of Ecosystem Studies	2016

PUBLICATIONS

h-index: 13, citations: 550 (as of 10/2026, Google Scholar)

Manuscripts in Preparation

Yang, L.H., Zhang Y., Li C., Shen S., van Donkelaar A., Zhang D., Bell M.L., Martin R.V. Changing Air Quality Response to Emission Controls in the US. *In Prep.*

Yang, L.H., Bowman K.W., Kim H., Miyazaki K. Megacities as Indicators of National NO_x and CO_2 Emission Trends. *In Prep.*

First-Author Papers

Yang, L.H., Jacob, D.J., Bates, K.H., Lin, H., Allen H.M., Müller J.F., Brown, S.S., Dang, R., Colombi N.K., Zhai, S., Yantosca, R.M., Brewer, J.F., Ng, N.L., Crounse, J.D., Wennberg, P.O., Li, K., Liao, H. Modeling of Methyl Hydroperoxide Observations in Urban and Remote Air over South Korea:

Methylperoxy Radical Chemistry and Inference of Atmospheric Methanediol. *Geophysical Research Letters*, 52(22), e2025GL118267, 2025. https://doi.org/10.1029/2025GL118267

Yang, L.H., Jacob, D.J., Lin, H., Dang, R., Bates, K.H., East, J.D., Travis, K.R., Pendergrass, D.C., Murray, L.T. Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. *Geophysical Research Letters*, 52(5), e2024GL112445, 2025. https://doi.org/10.1029/2024GL112445

Yang, L.H., Jacob, D.J., Dang, R., Oak, Y.J., Lin, H., Kim, J., Zhai, S., Colombi, N.K., Pendergrass, D.C., Beaudry, E., Shah, V., Feng, X., Yantosca, R.M., Chong, H., Park, J., Lee, H., Lee, W.-J., Kim, S., Kim, E., Travis, K.R., Crawford, J.H., Liao, H. Interpreting Geostationary Environment Monitoring Spectrometer (GEMS) Geostationary Satellite Observations of the Diurnal Variation in Nitrogen Dioxide (NO_2) over East Asia. *Atmospheric Chemistry and Physics*, 24(12), 7027–7039, 2024. https://doi.org/10.5194/acp-24-7027-2024

Yang, L.H., Jacob, D.J., Colombi, N.K., Zhai, S., Bates, K., Shah, V., Beaudry, E., Yantosca, R.M., Lin, H., Brewer, J., Chong, H., Travis, K., Crawford, J., Lamsal, L., Koo, J.-H., Kim, J. Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. *Atmospheric Chemistry and Physics*, 23(4), 2465–2481, 2023. https://doi.org/10.5194/acp-23-2465-2023

Yang, L.H., Hagan, D.H., Rivera-Rios, J.C., Kelp, M.M., Cross, E.S., Peng, Y., Kaiser, J., Williams, L.R., Croteau, P.L., Jayne, J.T., Ng, N.L. Investigating the Sources of Urban Air Pollution Using Low-Cost Air Quality Sensors at an Urban Atlanta Site. *Environmental Science & Technology*, 56(11), 7063–7073, 2022. https://doi.org/10.1021/acs.est.1c07005

Yang, L.H., Takeuchi, M., Chen, Y., Ng, N.L. Characterization of Thermal Decomposition of Oxygenated Organic Compounds in FIGAERO-CIMS. *Aerosol Science and Technology*, 55(12), 1321–1342, 2021. https://doi.org/10.1080/02786826.2021.1945529

Yang, L.H., Han, B.A. Data-driven Predictions and Novel Hypotheses about Zoonotic Tick Vectors from the Genus Ixodes. *BMC Ecology*, 18(7), 2018. https://doi.org/10.1186/s12898-018-0163-2

Co-Authored Papers

Lucille G.G., **Yang L.H.**, Mickley L.J., Oak Y.J. Comparing rural and urban NO_2 column densities from GEMS across East Asia: Diurnal and seasonal differences. *Atmospheric Environment*, 382, 122176, 2026. https://doi.org/10.1016/j.atmosenv.2026.122176

Dang R., Jacob D.J., Wang H., Nowlan C.R., Abad G.G., Chong H., Liu X., Shah V., **Yang L.H.**, Oak Y.J., Marais E.A., Horner R.P., Rollins A.W., Crawford J.H., Li K., and Liao H. High-resolution Geostationary Satellite Observations of Free Tropospheric NO_2 over North America and Implications for Lightning Emissions. *Proceedings of the National Academy of Sciences*, 122(42), e2510535122, 2025. https://doi.org/10.1073/pnas.2510535122

Colombi N.K., Jacob D.J., Ye X., Yantosca B., Bates K., Pendergrass D.C., **Yang L.H.**, Li K., and Liao H. Large and Increasing Stratospheric Contribution to Tropospheric Ozone over East Asia. *Atmospheric Chemistry and Physics*, 26(4), 2623–2633, 2026. https://doi.org/10.5194/acp-26-2623-2026

Pendergrass, D.C., Jacob, D.J., Oak, Y.J., Dang, R., **Yang, L.H.**, Beaudry, E., Colombi, N.K., Zhai, S., Kim, H., Choi, J.S., Park, J.S., Kim, S., Li, K., Liao, H. Wintertime Trends of Fine Particulate Matter ($PM_{2.5}$) in South Korea, 2012–2022: Response of Nitrate and Organic Components to Decreasing NO_x Emissions. *Geophysical Research Letters*, 52(19), e2025GL116091, 2025. https://doi.org/10.1029/2025GL116091

Westgate, S., Shivji, Z., **Yang, L.H.**, Rivera-Rios, J.C., Ng, N.L. Field evaluation of low-cost particulate matter (PM) sensor instruments in indoor and outdoor environments: A university lecture hall and urban southeastern United States. *Aerosol Science and Technology*, 59(10), 1250–1271, 2025. https://doi.org/10.1080/02786826.2025.2484244

Beaudry, E., Jacob, D.J., Bates, K., Zhai, S., **Yang, L.H.**, Pendergrass, D.C., Colombi, N., Simpson, I., Wisthaler, A., Hopkins, J., Ke, L., Liao, H. Ethanol and Methanol in South Korea and China: Evidence for Large Anthropogenic Emissions Missing from Current Inventories. *ACS ES&T Air*, 2(4), 456–465, 2025. https://doi.org/10.1021/acsestair.4c00210

Dang, R., Jacob, D.J., Zhai, S., **Yang, L.H.**, Pendergrass, D.C., Coheur, P., Clarisse, L., Van Damme, M., Choi, J., Park, J., Liu, Z., Xie, P., Liao, H. A Satellite-Based Indicator for Diagnosing Particulate Nitrate Sensitivity to Precursor Emissions: Application to East Asia, Europe, and North America. *Environmental Science & Technology*, 58(45), 20101–20113, 2024. https://pubs.acs.org/doi/10.1021/acs.est.4c08082

Lin, H., Emmons, L.K., Lundgren, E.W., **Yang, L.H.**, Feng, X., Dang, R., Zhai, S., Tang, Y., Kelp, M.M., Colombi, N.K., Eastham, S.D., Fritz, T.M., Jacob, D.J. Intercomparison of GEOS-Chem and CAM-chem Tropospheric Oxidant Chemistry within the Community Earth System Model Version 2 (CESM2). *Atmospheric Chemistry and Physics*, 24(15), 8607–8624, 2024. https://doi.org/10.5194/acp-24-8607-2024

Oak, Y.J., Jacob, D.J., Balasus, N., **Yang, L.H.**, Chong, H., Park, J., Lee, H., Lee, G.-T., Ha, E.-J., Park, R.J., Kwon, H.-A., Kim, J. A Bias-Corrected GEMS Geostationary Satellite Product for Nitrogen Dioxide Using Machine Learning to Enforce Consistency with the TROPOMI Satellite Instrument. *Atmospheric Measurement Techniques*, 17(17), 5147–5159, 2024. https://doi.org/10.5194/amt-17-5147-2024

Varon, D., Jervis, D., Pandey, S., Gallardo, S., Balasus, N., **Yang, L.H.**, Jacob, D. Quantifying NO_x Point Sources with Landsat and Sentinel-2 Satellite Observations of NO_2 Plumes. *Proceedings of the National Academy of Sciences*, 121(27), e2317077121, 2024. https://doi.org/10.1073/pnas.2317077121

Westervelt, D., Isavulambire, P., Yombo Phaka, R., **Yang, L.H.**, Raheja, G., Milly, G., Selenge, J.-L., Mulumba, J.P., Bousiotis, D., Djibi, B., McNeill, V.F., Ng, N.L., Pope, F., Mbela, G., Konde, J. A Low-Cost Investigation into Sources of $PM_{2.5}$ in Kinshasa, DRC. *ACS ES&T Air*, 1(1), 43–51, 2023. https://doi.org/10.1021/acsestair.3c00024

Zhai, S., Jacob, D.J., Pendergrass, D., Colombi, N., Shah, V., **Yang, L.H.**, Zhang, Q., Sun, Y., Wang, S., Luo, G., Yu, F., Woo, J.-H., Kim, Y., Kim, H., Dibb, J.E., Lee, T., Han, J.-S., Li, K., Liao, H. Coarse Particulate Matter Air Quality in East Asia: Implications for Fine Particulate Nitrate. *Atmospheric Chemistry and Physics*, 23(7), 4271–4281, 2023. https://doi.org/10.5194/acp-23-4271-2023

Colombi, N.K., Jacob, D.J., **Yang, L.H.**, Zhai, S., Shah, V., Grange, S.K., Yantosca, R.M., Kim, S., Liao, H. Why Is Ozone in South Korea and the Seoul Metropolitan Area So High and Increasing? *Atmospheric Chemistry and Physics*, 23(7), 4031–4044, 2023. https://doi.org/10.5194/acp-23-4031-2023

REPORT

Yang, L.H., Kim J, Ahn D.H., 2021. Air pollution: an introduction to its causes, effects, and solutions. Ministry of Foreign Affairs of the Republic of Korea, National Council on Climate and Air Quality of the Republic of Korea. Government Publications Registration Number: 11-1262000-000285-01.

PRESENTATIONS (all as presenting author)

Invited Talks

2026 Understanding Atmospheric Composition Using Atmospheric Chemistry Modeling and Observations. **Caltech Engineering and Applied Sciences Trailblazers Symposium and Environmental Science and Engineering Department Seminar**, Pasadena, CA, October.

2026 Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. **Methane Emissions Technology Alliance (META) led by Stanford University and Lawrence Berkeley National Laboratory**, Remote, February.

2025 Modeling of Methyl Hydroperoxide Observations in Urban and Remote Air over South Korea: Methylperoxy Radical Chemistry and Inference of Atmospheric Methanediol. **Caltech Laboratory for the Study of Atmospheric Composition Group Meeting**, Pasadena, CA, USA, June.

2023 Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO_2) over East Asia. **NASA and EPA Joint Atmospheric Chemistry Group Meeting**, Remote, December.

2023 Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. **Atmospheric Measurement Group Meeting, Center for Astrophysics – Harvard & Smithsonian**, Cambridge, MA, USA, December.

Other Talks

2026 Evolving Sensitivity of $PM_{2.5}$ to Emission Controls in a Changing Atmospheric Chemical Regime. **The 12th International GEOS-Chem Meeting**, St. Louis, MO, June.

2025 Megacities as Indicators of National NO_x and CO_2 Emission Trends. **AGU Annual Meeting**, New Orleans, LA, USA, December.

2025 Advancing Chemical Transport Modeling for Air Quality, Satellite Retrievals, and the Clean Energy Transition. **Center for Aerosol and Engineering (CASE) seminar**, St. Louis, MO, USA, November.

2025 Co-evolution of Fossil Fuel CO_2 and NO_x in Megacities around the World. **NASA JPL Earth Action for Air Quality Meeting**, La Cañada Flintridge, CA, USA, May.

2025 Advancing Chemical Transport Modeling for Air Quality, Satellite Retrievals, and the Clean Energy Transition. **NASA JPL Tropospheric Composition Group Meeting**, La Cañada Flintridge, CA, USA, May.

2025 Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. **EGU Annual Meeting**, Vienna, Austria, April.

2025 Modeling of methyl hydroperoxide observations in urban and remote air over South Korea: Demonstration of high-NO and low-NO regimes for hydrocarbon oxidation, and inference of atmospheric methanediol. **AMS Annual Meeting**, New Orleans, LA, USA, January.

2024 Assessment of Hydrogen's Climate Impact Is Affected by Model OH Biases. **AGU Annual Meeting**, Washington, DC, USA, December.

2023 Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO_2) over East Asia. **The 14th GEMS Science Team Meeting**, Jeju, South Korea, September.

2022 Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. **The 13th GEMS Science Team Meeting**, Seoul, South Korea, November.

2022 Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. **The 10th International GEOS-Chem Meeting**, St. Louis, MO, June.

2021 Understanding the Sources of Urban Air Quality Using Low-Cost Air Quality Sensors. **The 39th AAAR Conference**, Remote, October.

Posters

2024 Interpreting GEMS Geostationary Satellite Observations of the Diurnal Variation of Nitrogen Dioxide (NO_2) over East Asia. **The 104th AMS Annual Meeting**, Baltimore, MD, USA, January.

2023 Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. **Joint Science Meeting for TEMPO, GeoXO, & TOLNet**, Huntsville, AL, USA, May.

2022 Tropospheric NO_2 Vertical Profiles over South Korea and Their Relation to Oxidant Chemistry: Implications for Geostationary Satellite Retrievals and the Observation of NO_2 Diurnal Variation from Space. **AGU Fall Meeting**, Chicago, IL, USA, December.

2019 Thermal Decomposition Characterization of Filter Inlet for Gases and AEROsols (FIGAERO) coupled with Chemical Ionization Time-of-Flight Mass Spectrometer (ToF-CIMS). **The 37th AAAR Conference**, Portland, OR, October.

MENTORSHIP

Jaden Southern (Stanford University, class of 2026) Summer 2024
Co-mentor with Ruijun Dang, Yujin Oak, and Loretta Mickley.
Project: Evaluating TEMPO Satellite HCHO and NO_2 Products Against Ground-Based Pandora Instruments.

Lucy Gagnon (Williams College, Class of 2024; now Ph.D. student at Duke University) Summer 2023
Co-mentor with Yujin Oak and Loretta Mickley.
Project: Spatial and Temporal Differences in NO_2 Column Densities and Implications for Geostationary Satellite Product Applications Across Asia.

TEACHING EXPERIENCE

EPS 200: Atmospheric Chemistry and Physics Spring 2023, Spring 2024
Teaching Fellow, Harvard University Cambridge, MA

CEE 2300: Environmental Engineering Principles Spring 2018, Spring 2019
Teaching Assistant, Georgia Institute of Technology Atlanta, GA

ME 3322: Thermodynamics Spring 2019
Grader, Georgia Institute of Technology Atlanta, GA

SERVICE

Reviewer	*Atmospheric Measurement Techniques; npj Clean Air; Environmental Science: Atmospheres* NASA review panel (2025, 2026)
Leader	Group Life and Community Subgroup, Harvard Atmospheric Chemistry Modeling Group (ACMG) (2024–2025)
Organizer	Journal Club, Harvard ACMG (2022–2024) Graduate Student Forum, Harvard ACMG (2022–2024) Undergraduate Research Symposium 2024, Harvard ACMG (2024)
Member	American Geophysical Union, American Meteorological Society, Sigma Xi
Volunteer	Science Club for Girls, Boston (2021)